

Session Objectives

1. Explain the difference between primary and secondary immunodeficiency (ID) and outline some causes of secondary ID.
2. Explain the signs and symptoms that should raise suspicion for a primary ID (PID).
3. Identify the history and physical exam components important when considering an ID.
4. For the following PIDs, explain the underlying immune defect, the infectious clinical features, the non-infectious clinical features, laboratory features and the main treatment approaches:
 - Common Variable Immunodeficiency
 - Selective IgA Deficiency
 - IL-12 Receptor Deficiency
 - Chronic Mucocutaneous Candidiasis
 - Chédiak-Higashi Disease
 - Chronic Granulomatous Disease

Primary vs Secondary Immunodeficiency (ID)

**Definition: defect in the immune system resulting in a weakened
OR dysregulated immune defense**

Usually presents as unusual, recurrent or prolonged/severe infection

Primary ID:

- Inherent Defect in Immunity
- Most arise from inborn deviation in genetic code
- Usually presents in infancy or childhood, rarely adulthood
- Problems with B cells are most common PID (>50%)

Secondary ID:

- Medications – immunosuppressants, systemic steroids
- Viral infections – HIV, measles, rubella
- Protein losing states – nephrotic syndrome
- Metabolic disorders – diabetes, malnutrition, vitamin deficiency
- Other: prematurity,, acquired asplenia, acquired neutropenia from drugs or virus, autoimmune disorder (ex. SLE), malignancy

B Cell Disorders

- Usually no signs or symptoms until 3-4 months of age due to maternal IgG protection
- Antibodies are essential in sinopulmonary and mucosal surface protection
- Commonly present as recurrent otitis media, sinusitis or pneumonia but can have osteomyelitis, meningitis or sepsis
- Classic pathogens:
 - *Streptococcus pneumonia*, *Haemophilus influenza*, *Staph aureus*, *Pseudomonas aeruginosa*, *Mycoplasma* species
 - Chronic diarrhea from *Giardia lamblia*, *Salmonella enterica*, *Campylobacter jejuni*

B Cell

**Common Variable
Immunodeficiency**

Selective IgA Deficiency

X linked Agammaglobulinemia

Common Variable Immunodeficiency (CVID)

Defect: Heterogeneous group of disorders of Impaired Antibody Production

- Phenotypically normal B-cells do not differentiate due to an arrest in plasma cells differentiation
 - Results in hypogammaglobulinemia: Low IgG, Low IgA +/- Low IgM,
 - B cell numbers can be normal or low
- Various genetic defects and no specific genetic defect identified in most patients
- Defects can be due to AD, AR, or sporadic heritability
- Equally in males and females
- Usually presents in 2nd – 3rd Decade

Common Variable Immunodeficiency (CVID)

Clinical Features:

Infectious

- Recurrent sinopulmonary infections: *Streptococcus pneumoniae*, *Haemophilus influenza*, *Mycoplasma*
- Chronic GI infections or malabsorption: *Giardia*, *Campylobacter*, *Salmonella*, *Helicobacter*, *Enterovirus*
- Can also have septic arthritis, meningitis or sepsis

Non-infectious

- **Features of Immune Dysregulation**
 - Autoimmune disease: RA, thyroiditis, autoimmune hemolytic anemia, alopecia, vitiligo, pernicious anemia
 - Malignancy: Increased risk of lymphoma (usually EBV-associated)
 - Granulomatous Disease: GI, liver, lungs
- Chronic Lung Disease: bronchiectasis, interstitial lung disease
- GI Disorders: IBD, protein-losing enteropathy

Selective IgA Deficiency

Defect: Isolated absence (or near absence) of serum/secretory IgA (<5-10 mg/dL).

- Underlying defect is unknown
- Phenotypically normal B cells

Clinical Features: Variable but Majority are asymptomatic,

The Most common Primary ID found in approximately 1 in 500 people

- **Infectious:**

- Recurrent sinopulmonary infections: *Strep. pneumoniae*, *H. influenzae*
- GI infections: *Giardia lamblia*

- **Non-infectious:**

- Anaphylactic reaction to blood products/IVIg
- Food or Respiratory allergies
- GI disorders (Celiac disease, inflammatory bowel disease)
- Increased incidence of auto-immune disorders

Question 1

A 5-year-old female presents to your clinic with watery diarrhea and stomach cramps that have been present for multiple days. She has a PMH of a sinus infection and multiple GI problems within the past year and is up to date on immunizations with no adverse reactions reported. Stool antigen testing is positive for Giardia, and she is admitted to the hospital for IV fluids and recovery.

Once recovered, she returns to your office for follow up testing. A CBC and Ig panel show IgA of <5 mg/dL and normal IgM and IgG. What is the most likely diagnosis?

- A. Common Variable Immunodeficiency
- B. X-linked Agammaglobulinemia
- C. Selective IgA Deficiency
- D. Chronic Granulomatous Disease



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T Cell Disorders

- Result in both Humoral and Cellular Immune Defects
 - Both are because T-cells and Antigen presenting cells provide necessary signals for B-cell differentiation
- Susceptible to bacterial, fungal, viral, mycobacterial and protozoal infections
 - Similar bacteria as with antibody deficiency
 - *Pneumocystis jiroveci*, *Mycobacteria*, *Cryptosporidium parvum*, *Candidia*, *Aspergillus*
 - Viral infections often more severe, persistent, and disseminated

T Cell

Job Syndrome
(AD Hyper-IgE
Syndrome)

Thymic Aplasia

**IL-12 Receptor
Deficiency**

**Chronic
Mucocutaneous
Candidiasis**

Question 2

A 4-year-old girl presents to the clinic with her mother for a recent nail infection that has not gotten any better with over-the-counter creams. She has a history of multiple infections of her skin and mouth, which usually respond to treatment with an antimicrobial cream that her mom cannot remember the name of. A picture of her nails is shown below. Which of the following is the most likely cause for this patient's symptoms?

- A. Chediak-Higashi Syndrome
- B. Chronic Mucocutaneous Candidiasis
- C. Common Variable Immunodeficiency
- D. IL-12 Receptor Deficiency



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Phagocytic Dysfunction Disorders

- Phagocytic impairment can be due to inadequate neutrophil quantity or function
 - Neutrophils play an important role in the clearance of bacterial and fungal pathogens
- Primary ID due to defective phagocytic capacity is characterized by bacterial and fungal pathogens
 - Deep tissue infection, pneumonia, adenitis, sepsis, severe gingivitis, poor wound healing
 - *Staph aureus*, *Pseudomonas aeruginosa* can be life-threatening
- <20% of immunodeficiencies

Phagocytic

Leukocyte Adhesion
Deficiency Type 1

**Chediak-Higashi
Syndrome**

**Chronic granulomatous
disease**

Chédiak-Higashi Syndrome

Labs: Primary test is genetic testing for CHS1/LYST gene

- Peripheral smear can find giant cytoplasmic granules in leukocytes and platelets
- Neutropenia
- Often present with hypergammaglobulinemia due to frequent infections
- Abnormal platelet aggregation, bleed time

Treatment:

- Prophylactic antibiotics (Trimethoprim-Sulfamethoxazole)
- Aggressive treatment of infections
- IFN - γ helps with NK cell function
- G-CSF (Granulocyte colony stimulating factor) corrects neutropenia and decreases infection
- HSCT (hematopoietic stem cell transplant) is the treatment of choice
 - Will prevent accelerated phase
 - Does not prevent neurologic degeneration or oculocutaneous albinism

Chronic Granulomatous Disease (CGD)

Defect: Defective Phagocytes with inability to form respiratory burst to kill ingested catalase-positive bacteria or fungi

- Transmission: X-linked (65%) or Autosomal Recessive (35%)
 - Involve genes responsible for the NADPH oxidase complex
 - NADPH oxidase complex helps with respiratory/oxidative burst which is an important immune system process for helping phagocytes degrade internalized pathogens.
- In CGD, there is a decreased production of superoxide and H_2O_2 [respiratory burst] so microbes not killed

Clinical Features:

- Characterized by recurrent and severe bacterial and fungal infection with catalase + organism and granulomata formation
 - Think of organs that are barriers from infection [skin, GI tract, lungs, lymph nodes, spleen]
- Usually presents in first 5 years of life

Chronic Granulomatous Disease (CGD)

Infectious – usually focal but may see bacteremia and fungemia when local infections not treated appropriately

- Types of infections: Pneumonia, Lymphadenitis, Osteomyelitis, Skin
- Catalase-positive Bacterial organisms
 - *Staph aureus* is most common, *Serratia marcesens*, *Burkholderia cepacia*, *Nocardia*, *Pseudomonas*, *Salmonella*
- Catalase-positive Fungal organisms
 - *Aspergillus*, *Candida albicans*

Non-infectious – Growth failure, Anemia of chronic disease, and **Granulomas**

- Granulomas may be the presenting symptom and occur anywhere
- Pyloric obstruction, bladder outlet obstruction, rectal fistula, granulomatous colitis

Chronic Granulomatous Disease (CGD)

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Labs:

- Neutrophil Function Testing
 - Dihydrorhodamine (DHR)123 Test – Flow cytometry test; Sensitive and Quantitative Test
 - Measures oxidant production showing increased fluorescence when DHR is oxidized via NADPH oxidase pathway
 - Preferred as can distinguish between X-linked and autosomal forms
 - Nitroblue Tetrazolium (NBT) test: - Qualitative test
 - Simple and rapid determination of phagocyte NADPH oxidase activity.
 - NBT is reduced by superoxide and changes from yellow to dark blue in normal cells.
 - X-linked carriers can be identified with this test.
- Genetic Testing
- Other Lab Findings: Hypergammaglobulinemia, Elevated ESR or CRP, Anemia, Hypoalbuminemia

Treatment:

- Aggressively treat active infections: may be more difficult if deep-seated and abscesses may require surgical drainage
- Prophylactic treatment with Trimethoprim-sulfamethoxazole, Itraconazole, and IFN- γ
- HSCT (hematopoietic stem cell transplant) is curative
- Gene therapy is being researched

Question 4

A 21-year-old female presents with past medical history of autoimmune thrombocytopenia and recurrent sinusitis throughout childhood/adolescence, presents to for follow-up after a recent hospitalization for pneumonia. She has previously been hospitalized for pneumonia. On review of systems, she endorses chronic diarrhea and denies fever, chills and weight loss. X-ray on admission showed non-caseating granuloma in the R and L lungs. Laboratory analysis shows decreased IgG, IgM and IgA. What is the most likely diagnosis?

- A. Chronic Mucocutaneous Candidiasis
- B. Common variable Immunodeficiency
- C. Wiskott Aldrich Syndrome
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Question 4

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Summary

- Immunodeficiency disorders involve an impaired and dysregulated immune system, and can be caused by a variety of factors – both primary and secondary
- PIDs are associated with both infections and noninfectious features
- The infections associated with PIDs are recurrent, severe and/or characterized by unusual organisms
- Specific pathogens are associated with specific underlying immune defects; knowing these associations can give a clue to the underlying defect
- B cell disorders are associated with bacterial infections of the sinopulmonary tract
- T cell disorders are associated with opportunistic infections with bacteria, fungi and viruses
- Phagocytic disorders are associated with bacterial and fungal infections

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